

**Group Project Title:**

Dynamic Portfolio Allocation with Deep Deterministic Policy Gradient Project Report

**Group 10 | Group Code: DDPG-1**

**AI DECLARATION FORM**

**GROUP MEMBERS**

|                           |          |
|---------------------------|----------|
| 1. Samuel Maawo Akombasam | 22424856 |
| 2. Prince Obeng Nkoah     | 22424742 |
| 3. Douglas Adweeda        | 22425098 |

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**1. Tool Used**

Claude (Anthropic), accessed interactively via the Claude web interface, across an extended, multi-session conversation during the design, implementation and evaluation.

**2. Parts of the Work Where AI Was Used**

The table below states, plainly, which components of the submission were produced by the AI tool rather than independently written by the group.

| Component                                                                                    | AI Involvement | Detail                                                                             |
|----------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|
| MDP formulation<br>(state/action/reward/termination/discount,<br>Markov property discussion) | Drafted        | Course catalogue's candidate<br>state/action/reward variables for<br>Option DDPG-1 |
| Environment implementation<br>(env/portfolio_env.py)                                         | Drafted        | Half implementation, including<br>reward and termination logic                     |
| Training loop (train.py)                                                                     | Drafted        | Multi-seed training procedure                                                      |

**3. Declaration**

We confirm that the description above accurately reflects how generative AI was used in this project, and that we understand our individual and group responsibility for the correctness of everything submitted.

**Signed:** Samuel Maawo Akombasam

**Date:** 03/09/2026

**Signed:** Prince Obeng Nkoah

**Date:** 03/09/2026

**Signed:** Douglas Adweeda

**Date:** 03/09/2026